INCH-POUND

MIL-C-55302/181A
26 DEC 1989
SUPERSEDING
MIL-C-55302/181
15 April 1987

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES:
4-ROW PIN ASSEMBLIES, RIGHT ANGLE, 100 THROUGH 684
CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-C-55302.

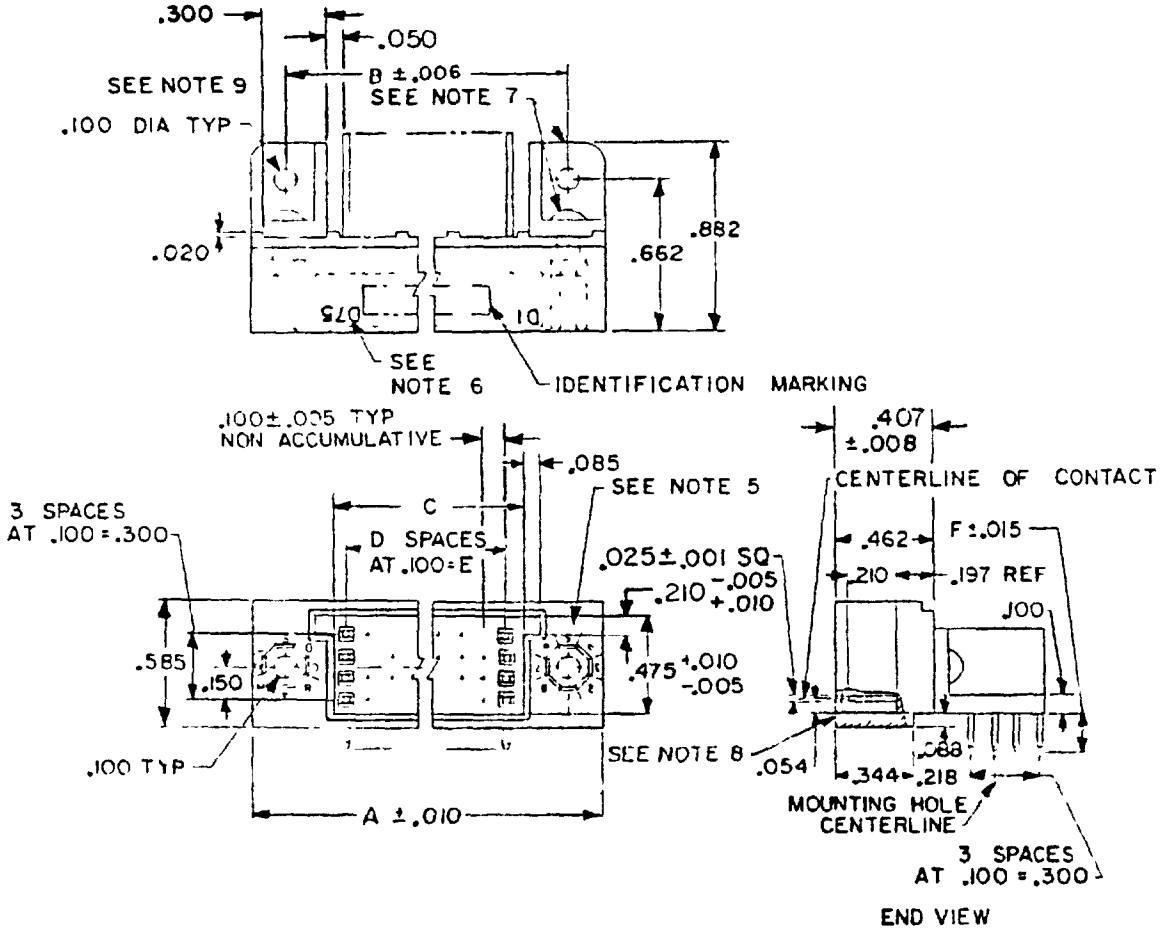
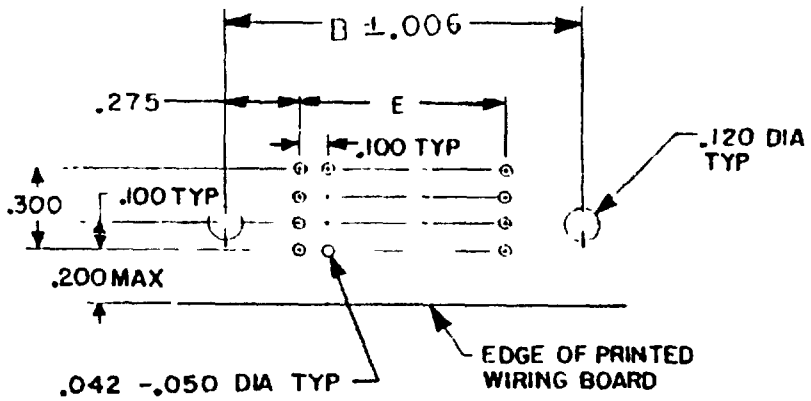


FIGURE 1 Connectors, receptacle assemblies .100 (2.54 mm) spacing, 100 through 300 contacts

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.



Recommended circuit layout

F ±.015 (0.38 mm)		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post (see note 9)
.120	3.05	IV	

Inches	mm	Inches	mm
.001	0 03	.150	3.81
.005	0.13	.197	5.00
.006	0.15	.200	5.08
.008	0.20	.210	5.33
.010	0.25	.218	5.54
.015	0.38	.275	6.98
.020	0.51	.300	7.62
.025	0.64	.344	8.74
.042	1.07	.407	10.34
.050	1.27	.462	11.73
.054	1.37	.475	12.06
.085	2.16	.585	14.86
.088	2.24	.662	16.81
.100	2.54	.882	22.40
.120	3.05		

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0 13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/180 and /185.
5. Location indicators embossed on surface
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Mounting ears and hardware are purchased separately. See MIL-C-55302/182 for mounting hardware.
8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ±5°.
9. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies, right angle, .100 (2.54 mm) spacing, 100 through 300 contacts - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.

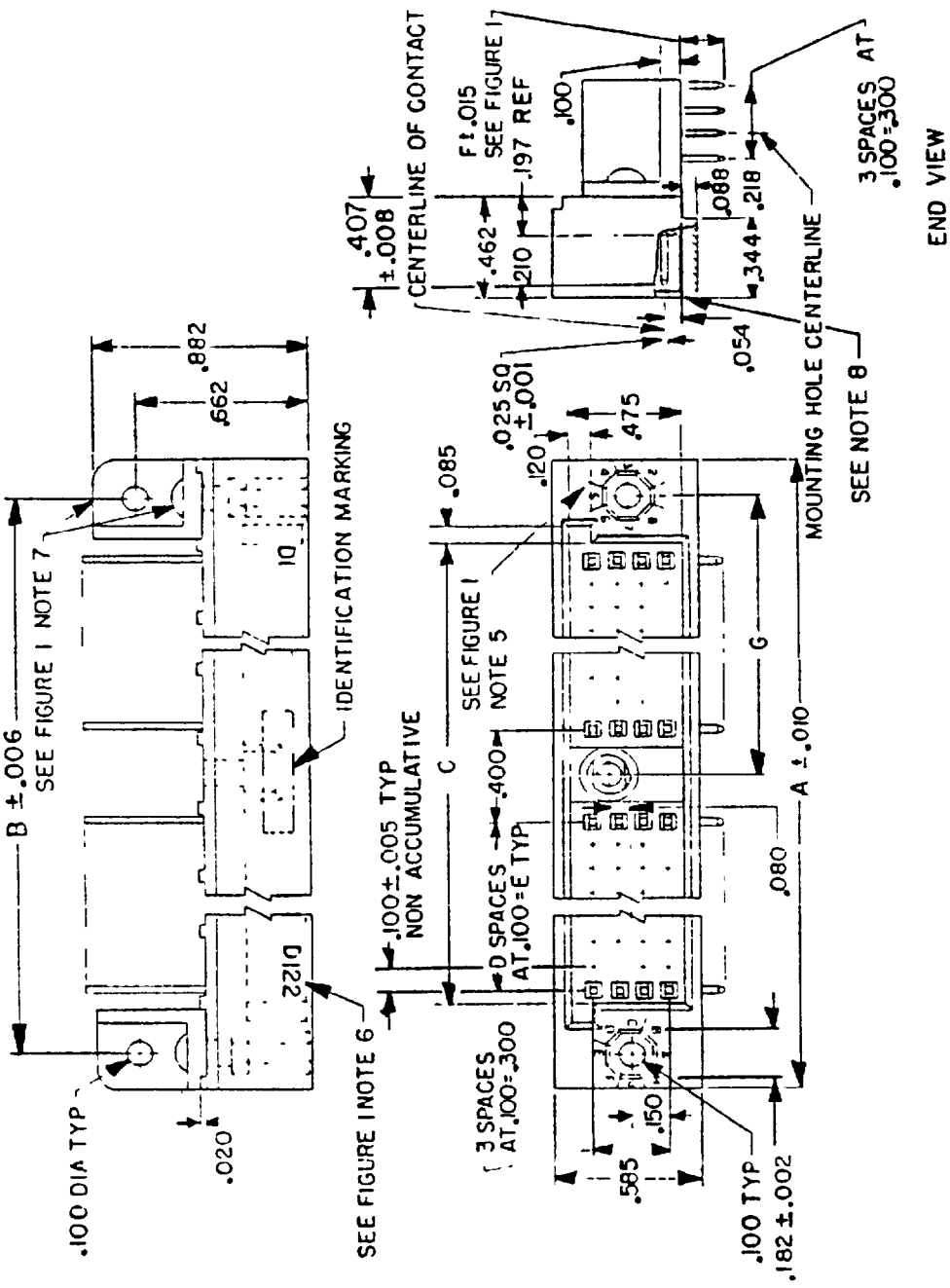
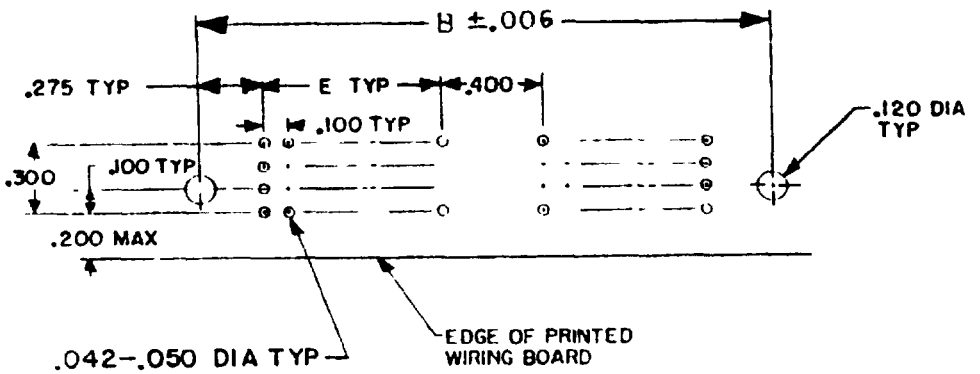


FIGURE 2. Connectors, pin assemblies, right angle, .100 (2.54 mm) spacing with one guide, 320 through 488 contacts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.



Recommended circuit layout

Inches	mm	Inches	mm
.001	0.03	.120	3.05
.002	0.05	.150	3.81
.005	0 13	.182	4.62
.006	0.15	.197	5.00
.008	0.20	.200	5.08
.010	0.25	.210	5.33
.015	0.38	.218	5.54
.020	0.51	.275	6.98
.025	0.64	.300	7.62
.042	1.07	.344	8.74
.050	1.27	.400	10.16
.054	1.37	.407	10.34
.080	2.03	.462	11.73
.085	2.16	.475	12.06
.088	2.24	.585	14.86
.100	2.54	.662	16.81
		.882	22.40

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2 Connectors, pin assemblies, right angle, 100 (2.54 mm) spacing with one guide, 320 through 488 contacts - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.

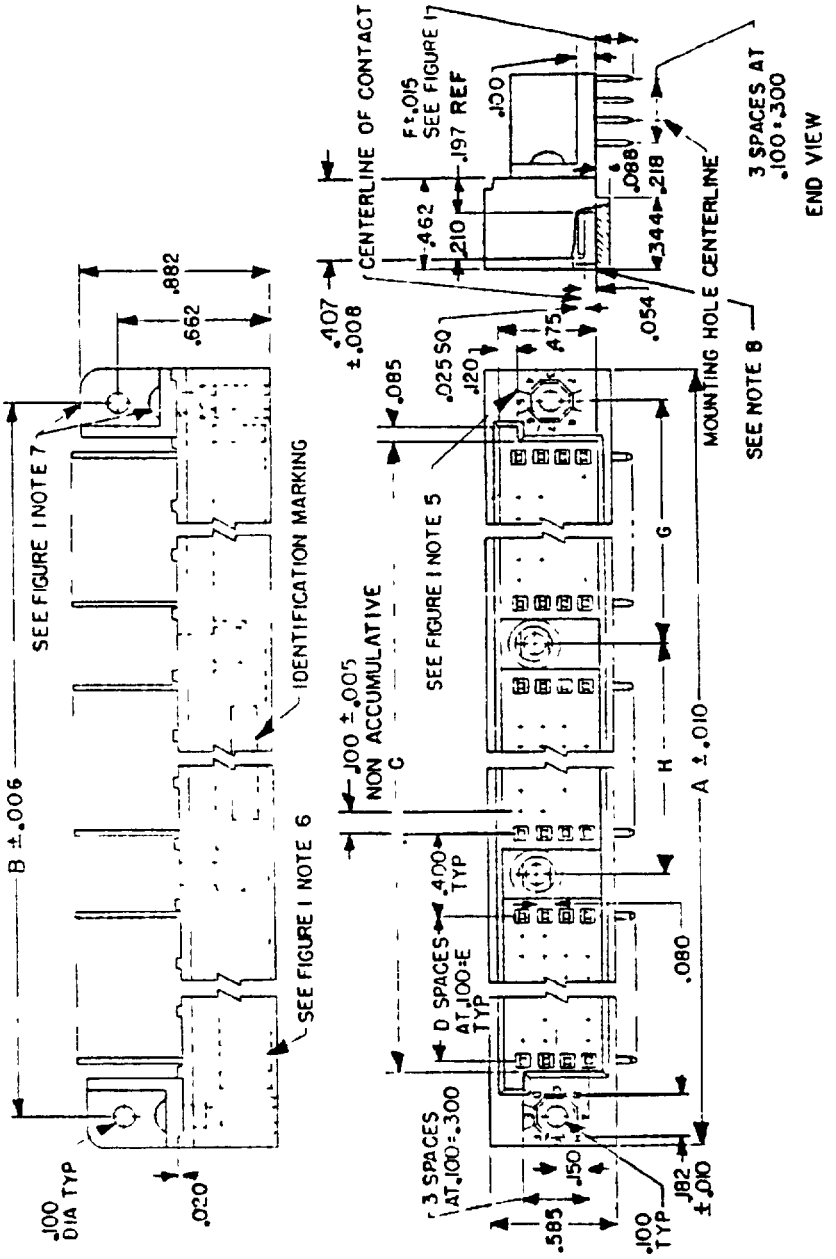
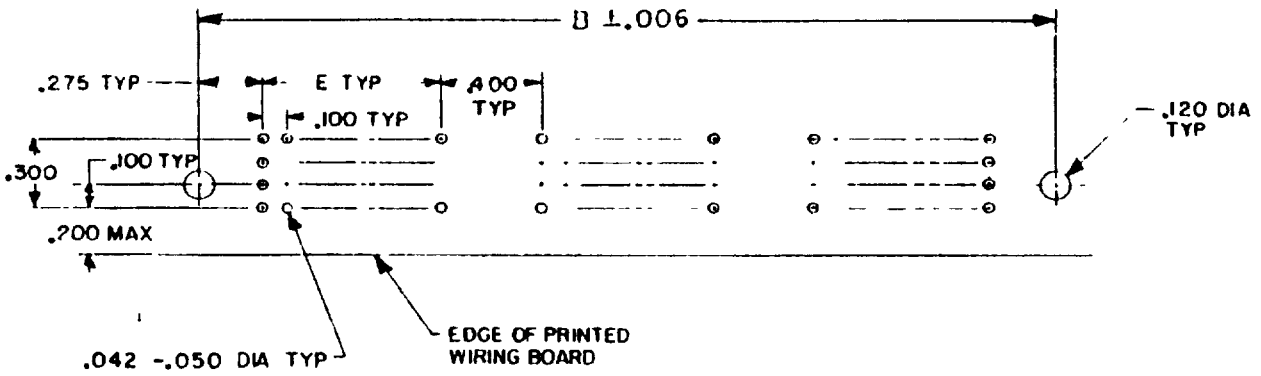


FIGURE 3. Connectors, pin assemblies, right angle, .100 (2.54 mm) spacing with two guides, 516 through 684 contacts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.



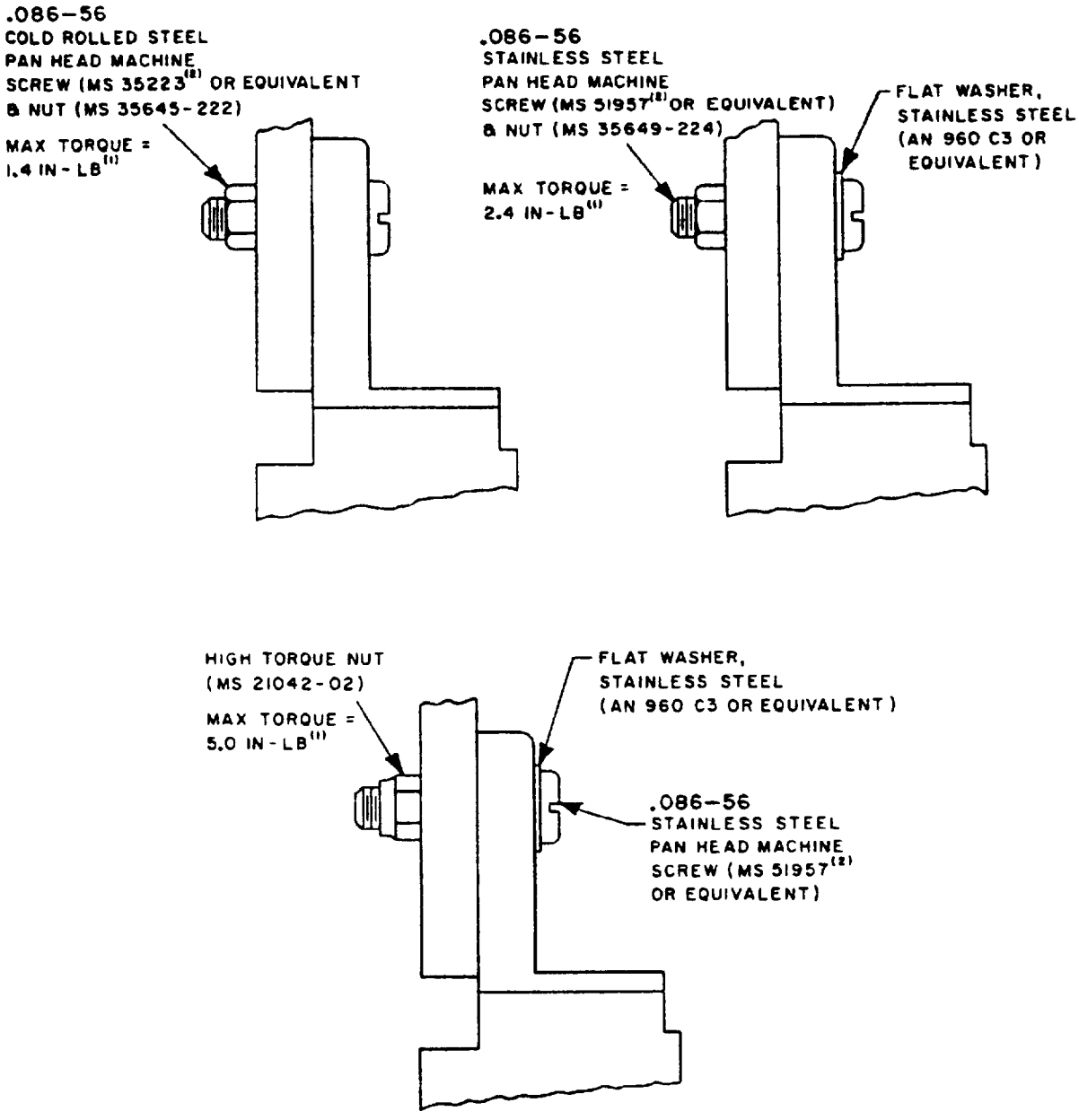
Recommended circuit layout

Inches	mm	Inches	mm
.005	0 13	.150	3.81
.006	0.15	.182	4 62
.008	0.20	.197	5.00
.010	0.25	.200	5.08
.015	0.38	210	5.33
.020	0.51	.218	5.54
.025	0.64	.275	6.98
.042	1.07	.300	7.62
050	1.27	.344	8.74
.054	1.37	.400	10.16
.080	2.03	.407	10.34
.085	2.16	.462	11.73
.088	2.24	.475	12.06
.100	2.54	.585	14.86
.120	3.05	.662	16.81
		.882	22.40

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3 Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4 With two guides, otherwise same notes as figure 1.
5 Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3. Connectors, pin assemblies, right angle, 100 (2 54 mm) spacing with two guides, 516 through 684 contacts - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.



NOTES
1 Torque wrench must be used when fastening connector to printed circuit board.
2 Length to be determined by user.

FIGURE 4. Mounting recommendations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 04	100	1	I IV	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)			1-25
09 12	128	1	I IV	3.950 (100.33)	3.650 (92.71)	3.250 (82.55)	31	3.100 (78.74)			1-32
17 20	160	1	I IV	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)			1-40
21 24	180	1	I IV	5.250 (133.35)	4.950 (125.73)	4.550 (115.57)	44	4.400 (111.76)			1-45
25 26	200	1	I IV	5.750 (146.05)	5.450 (138.43)	5.050 (128.77)	49	4.900 (124.46)			1-50
33 36	240	1	I IV	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
37 40	260	1	I IV	7.250 (184.15)	6.950 (176.53)	6.550 (166.37)	64	6.400 (162.56)			1-65
41 44	268	1	I IV	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)			1-67
45 48	280	1	I IV	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70
49 52	300	1	I IV	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)			1-75
57 60	344	2	I IV	9.650 (245.11)	9.350 (237.49)	8.950 (227.33)	42	4.200 (106.68)	4.675 (118.75)		1-86
69 72	416	2	I IV	11.450 (290.83)	11.150 (283.21)	10.750 (273.05)	51	5.100 (129.54)	5.575 (141.61)		1-104
81 84	488	2	I IV	13.250 (336.55)	12.950 (328.93)	12.550 (318.77)	60	6.000 (152.40)	6.475 (164.46)		1-222

See footnote at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number, number of contacts, contact type, and dimensions 1/ - Continued.

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
85 88	516	3	I IV	14.250 (361.95)	13.950 (354.33)	13.550 (344.17)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
89 92	540	3	I IV	14.850 (377.19)	14.550 (369.57)	14.150 (359.41)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
101 104	612	3	I IV	16.650 (422.91)	16.350 (415.29)	15.950 (405.13)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
105 108	636	3	I IV	17.250 (438.15)	16.950 (430.53)	16.550 (420.37)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1-159
113 116	684	3	I IV	18.450 (468.63)	18.150 (461.01)	17.750 (450.85)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

1/ Deleted dash numbers in the series 01 through 116 are canceled with no replacement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.

TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish. Contact engagement area shall have a minimum plating in accordance with MIL-C-55032. Solder post area (contact type I, .195 minimum length; type IV, .135 minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.

1/ Type C finish is deleted and superseded by type A or B.

REQUIREMENTS

Design and construction

Dimensions and configuration: See figures 1, 2, 3, 4 and table I.

Material In accordance with MIL-C-55302.

Plating See table II.

Current rating 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31)

Two keys, Part or Identifying Number (PIN) MS55302/31-04, and two .086-56 UNC-2A mounting screws, are recommended. Keys and screws shall be ordered separately.

Mating and unmating The maximum mating force in pounds shall be the number of contacts multiplied by .125; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing

Contact retention: 3 pounds, minimum

Dielectric withstanding voltage.

Sea level 900 volts rms

70,000 feet. 200 volts rms.

2 milliamperes maximum leakage current.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN) M55302/181 (and dash number from table I and type from table II)

Example M55302/181-01A.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 17

Review activities
Army - AR, MI
Navy - AS, SH
Air Force - 11, 85, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity
Air Force - 17

Agent
DLA - ES

(Project 5935-3715-09)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:56Z
Check the source to verify that this is the current version before use.